

Lecture 2: Linear Algebra

Originally written by Mauricio Caceres Bravo

Revised by Cole Davis

Webpage: cj-davis99.github.io

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Notation

- $\forall$ translates to “for all”
- $\exists$ translates to “there exists”
- $\mathbb{N} = \{1, 2, \dots\}$ is the set of natural numbers
- $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$ is the set of integers
- $\mathbb{Q} = \{p/q : p \in \mathbb{Z} \text{ and } q \in \mathbb{Z} \setminus \{0\}\}$ is the set of rational numbers
- $\mathbb{R}$ is the set of real numbers
- $\mathbb{C}$ is the set of complex numbers
- If S is a set and $n \in \mathbb{N}$, then S^n is the n^{th} order Cartesian product of S . E.g., $S^2 = S \times S$
- For any $\varepsilon > 0$, $B_\varepsilon(x)$ is the Euclidean ball around x with radius ε
- Unless otherwise specified, $d(x, y)$ is a metric on the contextual set x, y belong to
- The origin is always denoted as 0 regardless of the dimension of the space considered
- If A is a $N \times M$ matrix, $i \in \{1, \dots, N\}$, and $j \in \{1, \dots, M\}$, then $A_{-i, -j}$ is the submatrix of A with the i^{th} row and j^{th} column removed.

1 Motivation

In economics, it is very common to have **systems of equations**. Suppose we were studying the supply and demand of some good at price P . If we're interested in the equilibrium price and quantity for this good in the market, we may have the following simple set of equations:

$$\begin{aligned} Q_d &= a_0 + a_1 P \\ Q_s &= b_0 + b_1 P \\ Q_d - Q_s &= 0 \end{aligned} \tag{1}$$

We have an equation for quantity demanded (Q_d) at price P , an equation for quantity supplied (Q_s) at price P , and an equilibrium condition. We assume that a_0, a_1, b_0 , and b_1 are known market parameters.

In this setting, we solve for the equilibrium price and quantity by leveraging the equilibrium condition:

$$Q^s = Q^d \iff a_0 + a_1 P = b_0 + b_1 P \implies P = \frac{a_0 - b_0}{b_1 - a_1}$$

as long as $b_1 - a_1 \neq 0$. Solving for quantity yields,

$$Q^s = Q^d = a_0 + a_1 \frac{a_0 - b_0}{b_1 - a_1} = \frac{a_0 b_1 - a_0 a_1 + a_1 a_0 - a_1 b_0}{b_1 - a_1} = \frac{a_0 b_1 - a_1 b_0}{b_1 - a_1}$$

Linear algebra is a field of math that abstractly examines linear systems of equations (such as (1)) and delivers general methods and environments for which solutions exist and can be computed. Although (1) is a stylized market with linear supply and demand (which is, admittedly, a strong assumption), a thorough study of linear systems can be beneficial for helping us solve more general, nonlinear systems because solutions to these systems can often be approximated by solutions to linear approximations of the nonlinear system. Aside from solving systems of equations, linear algebra is also an incredibly helpful tool for formalizing high dimensional structures commonly encountered in math and economics.

2 Vector Spaces

Definition 1. A **field**, $\mathbb{F}$, is a set for which addition and multiplication are defined. Explicitly, $\mathbb{F}$ has the following properties:

1. $\forall x, y, z \in \mathbb{F}$

(a) $x + y \in \mathbb{F}$

(b) $x + y = y + x$

(c) $(x + y) + z = x + (y + z)$

2. $\forall x, y, z \in \mathbb{F}$

(a) $x \cdot y \in \mathbb{F}$

(b) $x \cdot y = y \cdot x$

(c) $(x \cdot y) \cdot z = x \cdot (y \cdot z)$

3. $\forall x, y, z \in \mathbb{F}, x \cdot (y + z) = x \cdot y + x \cdot z$
4. $\exists 0 \in \mathbb{F}$ such that $\forall x \in \mathbb{F}, x + 0 = x$ and $\exists (-x) \in \mathbb{F}$ such that $x + (-x) = 0$
5. $\exists 1 \in \mathbb{F}$ such that $\forall x \in \mathbb{F}, x \cdot 1 = x$ and, if $x \neq 0$, $\exists x^{-1} \in \mathbb{F}$ such that $x \cdot x^{-1} = 1$

Most linear algebra texts state all results assuming that the scalar field of interest is either $\mathbb{R}$ or $\mathbb{C}$. Most economics results take this a step further and assume that the scalar field of interest is $\mathbb{R}$. We will operate as the typical economist does and assume that the field of consideration is $\mathbb{R}$, but we state the definition of the next mathematical object in a general fashion:

Definition 2. A *vector space* V over a field $\mathbb{F}$ is a collection (or a set) of objects called vectors endowed with addition and scalar multiplication. Explicitly, V has the following properties:

1. **Additive closure:** $v + w \in V$ for any $v, w \in V$
2. **Commutativity:** $v + w = w + v$ for any $v, w \in V$
3. **Associativity:** $(v + w) + u = v + (w + u)$ for any $u, v, w \in V$.
4. **Additive identity:** $\exists 0 \in V$ s.t. $v + 0 = v$ for any $v \in V$ (the 0 vector or the origin).
5. **Additive inverse:** $\forall v \in V \exists -v \in V$ s.t. $v + (-v) = 0$.
6. **Multiplicative closure:** $\forall \alpha \in \mathbb{F}$ and $\forall v \in V, \alpha v \in V$
7. **Unit rule** or **multiplicative identity:** $1v = v$ for any $v \in V$.
8. **Multiplicative associativity:** $(\alpha\beta)v = \alpha(\beta v)$ for any $v \in V$ and $\alpha, \beta \in \mathbb{F}$.
9. **Distributivity:** For any $\alpha, \beta \in \mathbb{F}, v, w \in V$, we have

$$\alpha(v + w) = \alpha v + \alpha w \quad \text{and} \quad (\alpha + \beta)v = \alpha v + \beta v$$

Remark 1. If you are like me, you might think it's a bit odd that we are making a big deal of associativity, commutativity, and distributivity (like we did back in primary school). The reason is that a more formal treatment of linear spaces would not take any of these properties for granted, and would be very careful in discussing everything in this section (and henceforth) using just the definitions.

It can be shown that all finite dimensional vector spaces over $\mathbb{R}$ are equivalent to the vector space $\mathbb{R}^N$ for some $N \in \mathbb{N}$. This can be of course be proven formally (and is a fundamental result in any theoretical linear algebra class), but for this class I will simply discuss linear algebra results in $\mathbb{R}^N$.

Definition 3. Let V be a vector space and let $\{v_1, \dots, v_N\} \subseteq V$. A *linear combination* of $\{v_1, \dots, v_N\}$ is given by

$$\sum_{i=1}^N \alpha_i v_i = \alpha_1 v_1 + \dots + \alpha_N v_N$$

for some arbitrary set of coefficients $\{\alpha_1, \dots, \alpha_N\} \subseteq \mathbb{R}$.

Remark 2. A vector space is a space that contains all its linear combinations. That is, for any $\{v_i\}_{i=1}^N \subseteq V$ and $\{\alpha_i\}_{i=1}^N \subseteq \mathbb{R}$ we have

$$\sum_{i=1}^N \alpha_i v_i \in V$$

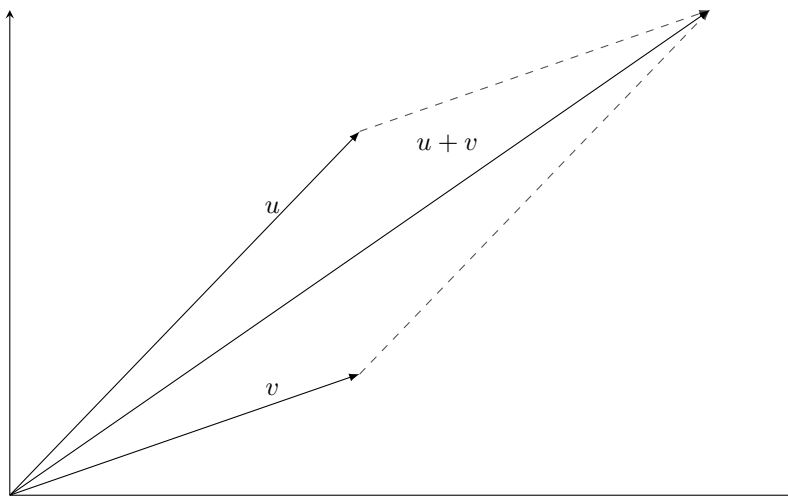


Figure 1: Example of Vector Addition

if and only if V is a vector space. This is not obvious (and in a more formal treatment we would prove it from the definitions), but it turns out one characterization of a vector space is a space that has all its linear combinations. Hence the reason why vector spaces are also known as **linear spaces**.

Definition 4. Let V be a vector space. We have $\{v_i\}_{i=1}^N \subseteq V$ is a **basis** of V if for every $u \in V$ there exists a *unique* linear combination of $\{v_i\}_{i=1}^N$ s.t.

$$u = \sum_{i=1}^N \alpha_i v_i$$

The coefficients $\{\alpha_i\}_{i=1}^N$ are called the **coordinates** of u in V with respect to the basis $\{v_i\}_{i=1}^N$.

The easiest example of a basis is the standard basis in $\mathbb{R}^N$:

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} \quad \dots \quad e_N = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Naturally any $u \in \mathbb{R}^N$ can be uniquely represented as a linear combination of the standard basis if we take $\alpha_i = u_i \forall i$; that is, if we set the coefficients equal to each of the entries in u .

Definition 5. Let V be a vector space; we say that $\{v_i\}_{i=1}^N \subseteq V$ is a **spanning set** of V if for every $u \in V$ there exists a linear combination of $\{v_i\}_{i=1}^N$ s.t.

$$u = \sum_{i=1}^N \alpha_i v_i$$

Note that in the definition above, the linear combination does not have to be unique. Hence every basis of a vector space V is a spanning set of V , but not every spanning set of V can be a basis. Hence the question: When is a spanning set a basis?

Definition 6. A linear combination with coefficients $\{\alpha_i\}_{i=1}^N$ is called **trivial** if $\alpha_i = 0$ for every i .

Definition 7. Let V be a vector space; we say that $\{\mathbf{v}_i\}_{i=1}^N \subseteq V$ are **linearly independent** iff the only linear combination of $\mathbf{v}_i$ that is equal to 0 is trivial. That is

$$\sum_i \alpha_i \mathbf{v}_i = 0 \implies \alpha_i = 0$$

Definition 8. Let V be a vector space; if $\{\mathbf{v}_i\}_{i=1}^N \subseteq V$ are not linearly independent, we say they are **linearly dependent**.

Theorem 1. Let V be a vector space; $\{\mathbf{v}_i\}_{i=1}^N \subseteq V$ is a basis for $V \iff \{\mathbf{v}_i\}_{i=1}^N$ are a linearly independent spanning set of V .

Claim 1. Let $\{\mathbf{v}_i\}_{i=1}^N, \{\mathbf{u}_i\}_{i=1}^M$ be any basis for a vector space V . Then $N = M$.

Proof. For the sake of contradiction, suppose $M > N$ (the $N < M$ case is a symmetric argument). We know that $\mathbf{u}_i \neq \mathbf{0}$ for all i (otherwise $\{\mathbf{u}_i\}_{i=1}^M$ is not linearly independent). Since $\{\mathbf{v}_i\}_{i=1}^N$ spans V ,

$$\mathbf{u}_1 = \sum_{i=1}^N \alpha_{1i} \mathbf{v}_i$$

where $\alpha_{1i^*} \neq 0$ for some $i^* \in \{1, \dots, N\}$. WLOG, label $\{\mathbf{v}_i\}_{i=1}^N$ so that $i^* = 1$. Then

$$\mathbf{v}_1 = -\sum_{i=2}^N \frac{\alpha_{1i}}{\alpha_{11}} \mathbf{v}_i + \frac{1}{\alpha_{11}} \mathbf{u}_1 \implies \mathbf{v}_1 \in \text{span}(\mathbf{v}_2, \dots, \mathbf{v}_N, \mathbf{u}_1) = V$$

Similarly,

$$\mathbf{u}_2 = \sum_{i=2}^N \alpha_{2i} \mathbf{v}_i + \alpha_{21} \mathbf{u}_1$$

If $\alpha_{2i} = 0$ for all $i \geq 2$, then $\{\mathbf{u}_1, \mathbf{u}_2\}$ is a linearly dependent set, a contradiction. Otherwise, we can, WLOG, suppose that $\alpha_{22} \neq 0$ so that

$$\mathbf{v}_2 = -\sum_{i=3}^N \frac{\alpha_{2i}}{\alpha_{22}} \mathbf{v}_i + \frac{\alpha_{21}}{\alpha_{22}} \mathbf{u}_1 + \frac{1}{\alpha_{22}} \mathbf{u}_2 \implies \mathbf{v}_2 \in \text{span}(\mathbf{v}_3, \dots, \mathbf{v}_N, \mathbf{u}_1, \mathbf{u}_2) = V$$

Repeat this procedure until we either find a $\mathbf{u}_m$ that can be written as a linear combination of $\{\mathbf{u}_1, \dots, \mathbf{u}_{m-1}\}$ – a contradiction – or until we’ve shown how to express $\mathbf{v}_N$ as a linear combination of $\{\mathbf{u}_1, \dots, \mathbf{u}_N\}$. In the latter case, we can then show that $\mathbf{u}_{N+1}$ can be written as a linear combination of $\{\mathbf{u}_1, \dots, \mathbf{u}_N\}$ because this set of vectors spans V . This is an immediate contradiction of the linear independence of $\{\mathbf{u}_i\}_{i=1}^M$ and completes the proof. $\square$

This claim establishes that bases of a vector space have the same cardinality. This prompts the following definition:

Definition 9. A vector space V has **dimension**, $\dim(V)$, equal to the number of elements in any of its bases.

3 Matrices

In this section and the remainder of these notes, we begin moving towards more practical matrix algebra that will be used throughout the first-year sequence. Fix $N \in \mathbb{N}$. We always default to assuming vectors in $\mathbb{R}^N$ are **column vectors**; that is,

$$\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_N \end{pmatrix}$$

We can turn $\mathbf{x} \in \mathbb{R}^N$ into a **row vector** by applying the **transpose** operation

$$\mathbf{x}' = \mathbf{x}^\top = \mathbf{x}^T = (x_1 \quad \dots \quad x_N) = (x_1, \dots, x_N)$$

An $M \times N$ matrix A is a collection of N M -dimensional vectors $\{\mathbf{a}_1, \dots, \mathbf{a}_N\} \subseteq \mathbb{R}^M$. For any N -dimensional vector $\mathbf{x} = (x_1, \dots, x_N)'$, we define

$$A\mathbf{x} = \sum_{i=1}^N x_i \mathbf{a}_i$$

Notice that $A\mathbf{x} \in \mathbb{R}^M$ and $A\mathbf{x}$ is not well-defined if $\mathbf{x}$ comes from a vector space whose dimension differs from the number of columns of A .

For the $M \times N$ matrix $A = (\mathbf{a}_1, \dots, \mathbf{a}_N)$, we define the transpose as

$$A^\top = \begin{pmatrix} \mathbf{a}_1^\top \\ \vdots \\ \mathbf{a}_N^\top \end{pmatrix}$$

Notice that $A^\top$ is now an $N \times M$ matrix.

Suppose A is an $M \times N$ matrix and B is an $N \times K$ matrix. We can define **matrix multiplication** as:

$$AB = A \begin{pmatrix} \mathbf{b}_1 & \dots & \mathbf{b}_K \end{pmatrix} = \begin{pmatrix} A\mathbf{b}_1 & \dots & A\mathbf{b}_K \end{pmatrix}$$

so AB is an $M \times K$ matrix. Unlike matrix multiplication, **matrix addition** is performed component wise, so it is only well-defined for matrices with identical row and column sizes. That is, if A and B are $M \times N$ matrices, then

$$A + B = \begin{pmatrix} \mathbf{a}_1 & \dots & \mathbf{a}_N \end{pmatrix} + \begin{pmatrix} \mathbf{b}_1 & \dots & \mathbf{b}_N \end{pmatrix} = \begin{pmatrix} \mathbf{a}_1 + \mathbf{b}_1 & \dots & \mathbf{a}_N + \mathbf{b}_N \end{pmatrix}$$

Some properties of matrix multiplication and addition for conformable matrices A, B, C that are important to be aware of:

1. **Associativity:** $(AB)C = A(BC)$ and $(A + B) + C = A + (B + C)$
2. **Distributivity:** $A(B + C) = AB + AC$
3. **Commutativity:** $A + B = B + A$

In general **matrix multiplication is not commutative**. That is, we cannot freely assume that $AB = BA$ (in fact, BA may not even be well-defined).

4 Matrix Inverse, Rank, and Determinant

Definition 10. The vector space spanned by the columns of a matrix A is the **column space** of A . The **rank** of a matrix, $\text{rank}(A)$, is the dimension of the column space (the maximum number of linearly independent columns).¹ A square matrix $N \times N$ is called **full-rank** if $\text{rank}(A) = N$ and **rank-deficient** if $\text{rank}(A) < N$.

Definition 11. A $N \times N$ matrix A is called **diagonal** if all its non-diagonal entries are 0.

Definition 12. The **identity matrix** is an $N \times N$ diagonal matrix with all diagonal entries equal to 1 (and non-diagonal entries equal to 0). For any $N \times N$ matrix A , $AI = IA = A$.

$$I = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

Definition 13. The $N \times N$ matrix A is said to be **left invertible** if $\exists C_L$ s.t. $C_L A = I$ the identity. We say it is **right invertible** if $\exists C_R$ s.t. $A C_R = I$. If A is left and right invertible with $C_L = C_R$, we say that A is **invertible** (or **non-singular**), and we denote $C_R = C_L = A^{-1}$ the **inverse** of A .²

Notice that, if A is left invertible, then it's injective. Further, if A is right invertible, then it's surjective. To see this, suppose A is left invertible and $A\mathbf{x} = A\mathbf{x}'$. Then $C_L A\mathbf{x} = C_L A\mathbf{x}' \implies \mathbf{x} = \mathbf{x}'$. Now, suppose that A ($M \times N$) is right invertible and $\mathbf{y} \in \mathbb{R}^M$. We know $\mathbf{y} = I\mathbf{y} = (A C_R)\mathbf{y} = A(C_R \mathbf{y})$.

Definition 14. If A is not invertible, we say that A is **singular**.

Theorem 2. Let A be a $N \times N$ matrix. A is invertible $\iff \text{rank}(A) = N$.

Some other properties of the rank:

- $\text{rank}(A) = \text{rank}(A^\top) = \text{rank}(A A^\top) = \text{rank}(A^\top A)$.
- $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$.
- $\text{rank}(CAB) = \text{rank}(A)$ if C, B are non-singular.

Example 1. Can we find the inverse of the matrix

$$B = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

¹It turns out that the row-rank of a matrix, the maximum number of linearly independent rows, is the same as the column-rank of a matrix. Hence we can just talk about the rank without clarifying row or column.

²The left and right inverses are the same for a square matrix if either exist, but this is a result, not an assumption. Further, note you can define the left and right inverses for non-square matrices, but such matrices needn't be invertible as only one of C_L or C_R might exist.

using elementary row operations?

$$\begin{bmatrix} 2 & 1 & & \\ 1 & 1 & & \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & & \\ & 1 & & \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1/2 & & \\ 1 & 1 & & \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1/2 & & \\ & 1 & & \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1/2 & & \\ 0 & -1/2 & & \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1/2 & & \\ & 1 & & \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & & \\ 0 & -1/2 & & \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & & \\ & 1 & & \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & & \\ 0 & 1 & & \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & & \\ 0 & 1 & & \end{bmatrix}$$

The steps here were:

1. Multiply the first row by $1/2$.
2. Multiply the second row by -1 and add the first row to the second row.
3. Add the second row to the first row.
4. Multiply the second row by -2 .

We have a shortcut for 2×2 matrices:

$$B = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad B^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

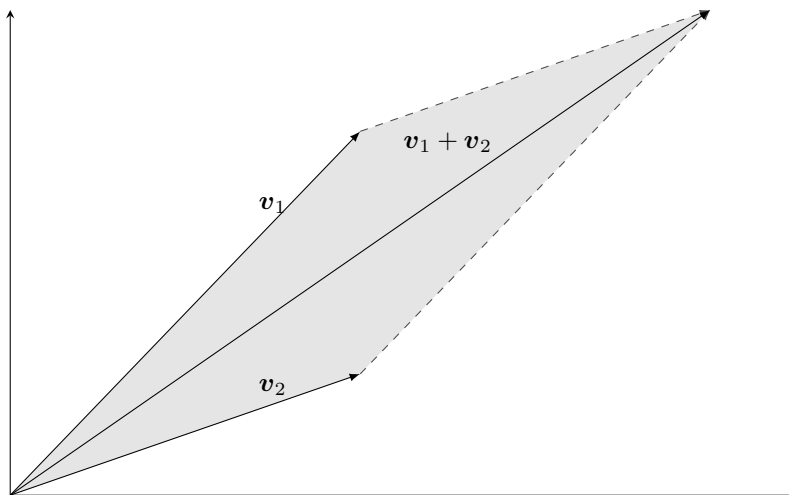
You can check this formula gives exactly what we found using row operations.

Definition 15. The *determinant* of a $N \times N$ square matrix A is

$$\det(A) = \sum_{j=1}^N (-1)^{i+j} a_{ij} \det(A_{-i,-j})$$

with i any fixed row of A and $A_{-i,-j}$ the sub-matrix of A that results from moving row i and column j .

Consider this picture:



The set

$$\{\mathbf{v} : \mathbf{v} = t_1 \mathbf{v}_1 + t_2 \mathbf{v}_2, \quad t_1, t_2 \in [0, 1]\}$$

will give the parallelogram, and the determinant of the 2×2 matrix $V = [\mathbf{v}_1 \ \mathbf{v}_2]$ will be the area of the parallelogram. (Rather, the absolute value of the determinant; if the determinant is negative that just says something about the direction of the vectors, but the intuition remains). Some properties of the determinant:

- If A and B are both square matrices, then $\det(AB) = \det(A) \det(B)$.
- $\det(A) = \prod_i a_{ii}$ if A is diagonal, upper triangular (i.e., only zeros entries below diagonal), or lower triangular (i.e., only zero entries above diagonal).
- $\det(I) = 1$ (immediate consequence of the point above).
- $\det(\alpha A) = \alpha^N \det(A)$ for any $\alpha \in \mathbb{R}$.
- If A is invertible, then $\det(A^{-1}) = \det(A)^{-1}$.

Another point to make about the determinant is that it is related to the rank, that is, whether the columns (or rows) of a square matrix A are linearly independent.

1. If A has a zero column or zero row, then $\det(A) = 0$.

To see this, note that from the formula of the determinant, if we have a row or column of all 0s, then each element of the sum will eventually be multiplied by 0, and the entire term will be 0. In the 2×2 case, if one of the vectors comprising a “side” of the parallelogram is the origin, the parallelogram will just be line which we know has no area.

2. If A has two columns or rows that are equal, then $\det(A) = 0$.

If two columns of A are equal, WLOG let them be the first two columns, consider the matrix P with

all diagonal entries equal to 1 and all but one non-diagonal entries equal to 0. Let $P_{21} = -1$. That is,

$$P = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ -1 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & & \vdots \\ \vdots & & & \ddots & \\ 0 & & & & 1 \end{bmatrix}$$

What is AP ? In this case, with $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_N]$, we have

$$AP = [\mathbf{a}_1 - \mathbf{a}_2 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_N] = [\mathbf{0} \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_N]$$

with the first column equal to 0 since we assumed the first two columns were equal. Finally, we found in the previous bullet that if a column was all 0s then the determinant was 0. Hence

$$0 = \det(AP) = \det(A) \det(P) \implies \det(A) = 0 \quad \text{or} \quad \det(P) = 0$$

Note $\det(P) = 1$, so it must be that $\det(A) = 0$.

3. If A has a columns that is a multiple of another column, then $\det(A) = 0$.

Again WLOG suppose these are the first two columns and let A be as above. If $\mathbf{a}_1 = c\mathbf{a}_2$ for some $c \in \mathbb{R}$, then we can define P almost identically, except $P_{21} = -c$. That is,

$$P = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ -c & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & & \vdots \\ \vdots & & & \ddots & \\ 0 & & & & 1 \end{bmatrix}$$

And again we have $AP = [\mathbf{a}_1 - c\mathbf{a}_2 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_N] = [\mathbf{0} \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_N]$. Hence $\det(AP) = \det(A) \det(P) = 0$, but again we have $\det(P) = 1$, so $\det(A) = 0$.

4. If A has a columns or rows that are linearly dependent, then $\det(A) = 0$.

This is really the point we wanted to make: A determinant of 0 is tied to the linear dependence of the columns of A . This follows from the work we did above: In this case, $\exists \{c_i\}$ s.t. $c_i \neq 0$ for some i s.t.

$$\mathbf{a}_j = \sum_{i \neq j} c_i \mathbf{a}_i$$

for some j . WLOG let $j = 1$, and the matrix P becomes

$$P = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ -c_2 & 1 & 0 & \cdots & 0 \\ -c_3 & 0 & 1 & & \vdots \\ \vdots & & & \ddots & \\ -c_N & & & & 1 \end{bmatrix}$$

So that

$$AP = \begin{bmatrix} \mathbf{a}_1 - c_2\mathbf{a}_2 - c_3\mathbf{a}_3 \cdots & \mathbf{a}_2 & \cdots & \mathbf{a}_N \end{bmatrix} = \begin{bmatrix} \mathbf{a}_1 - \sum_{i \neq 1} c_i \mathbf{a}_i & \mathbf{a}_2 & \cdots & \mathbf{a}_N \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{a}_2 & \cdots & \mathbf{a}_N \end{bmatrix}$$

So, one last time, $\det(AP) = \det(A)\det(P) = 0$ and $\det(P) = 1$ implies $\det(A) = 0$.

Theorem 3. A is rank-deficient $\iff \det(A) = 0$. Equivalently, A is non-invertible $\iff \det(A) = 0$.

Definition 16. For a $N \times N$ matrix A , the i, j **minor**, denoted as M_{ij} , is the *determinant* of the $(N - 1) \times (N - 1)$ sub-matrix obtained by deleting the i th row and j th column of A .

Definition 17. The i, j **cofactor** of the square $N \times N$ matrix A is given by

$$C_{ij} = (-1)^{i+j} M_{ij}$$

Thus we can rewrite Definition 15 as $\det(A) = \sum_{j=1}^N a_{ij} C_{ij}$ for any given i .

Definition 18. The **adjoint** of an $N \times N$ matrix A is the transpose of the cofactor matrix given by

$$\text{adj}(A) = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1N} \\ C_{21} & C_{22} & & \vdots \\ \vdots & & \ddots & \\ C_{N1} & C_{N2} & \cdots & C_{NN} \end{bmatrix}^T = \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{N1} \\ C_{12} & C_{22} & & \vdots \\ \vdots & & \ddots & \\ C_{1N} & C_{2N} & \cdots & C_{NN} \end{bmatrix}$$

Note the above is a matrix of scalars, since C_{ij} is the product of a determinant, which is a scalar, and $(-1)^{i+j}$, which is a scalar as well.

Theorem 4. If A is a non-singular $N \times N$ matrix then

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

For instance, for a 2×2 matrix we have

$$B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Applying the theorem,

$$B^{-1} = \frac{1}{\det(B)} \begin{bmatrix} C_{11} & C_{21} \\ C_{12} & C_{22} \end{bmatrix} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Theorem 5 (Cramer's Rule). *Let A be a $N \times N$ non-singular matrix s.t. $A\mathbf{x} = \mathbf{b}$. Then*

$$x_i = \frac{\det(\tilde{A}_i)}{\det(A)}$$

for

$$\tilde{A}_i = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_{i-1} & \mathbf{b} & \mathbf{a}_{i+1} & \cdots & \mathbf{a}_N \end{bmatrix}$$

That is, $\tilde{A}$ is the matrix obtained by replacing the i th column of A with $\mathbf{b}$.

Definition 19. The **trace** of an $N \times N$ matrix A is the sum of its diagonal elements,

$$\text{tr}(A) = \sum_{i=1}^N a_{ii}$$

Some properties of the trace:

- $\text{trace}(A + B) = \text{trace}(A) + \text{trace}(B)$
- $\text{trace}(AB) = \text{trace}(BA)$ if both products exist.
- $\text{trace}(\alpha A) = \alpha \text{trace}(A)$ for any $\alpha \in \mathbb{R}$.

An example which comes up in econometrics is that for any $N \times K$ matrix X s.t. $(X^T X)^{-1}$ exists, we have

$$\text{trace} \left(\underbrace{X}_A \overbrace{(X^T X)^{-1} X^T}^B \right) = \text{trace}((X^T X)^{-1} X^T X) = \text{trace}(I_{K \times K}) = K$$

Further,

$$\text{trace}(I_{N \times N} - X(X^T X)^{-1} X^T) = \text{trace}(I_{N \times N}) - \text{trace}(X(X^T X)^{-1} X^T) = N - K$$

Another useful trick of the trace that pops up in econometrics proofs is that $\alpha = \text{trace}(\alpha)$ when $\alpha \in \mathbb{R}$.

5 Eigenvalues and Eigenvectors

Take

$$A\mathbf{v} = \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

Note that for the standard basis,

$$\begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix} e_1 = 3e_1 \quad \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix} e_2 = -e_2$$

For this particular example, we can see the standard basis is exactly scaled by 3 and -1 . Hence, T acting on v scales v_1 by 3 and scales v_2 by -1 . Unfortunately, not all linear transformations have diagonal matrix

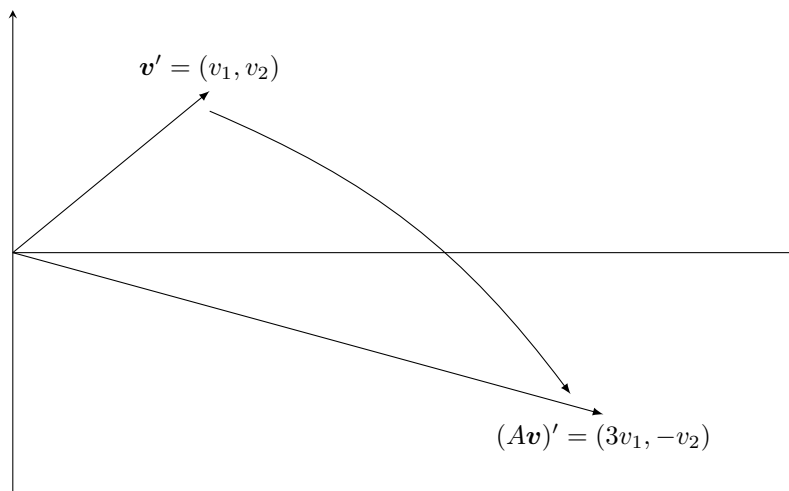


Figure 2: Visualization of a Transformation

representations that allow us to easily determine how they geometrically transform their arguments. This leads us to ask the following question: can we build a theory that helps us better understand the geometry of any given linear transformation?

Definition 20. Let A be a $N \times N$ square matrix. The $N \times 1$ vector $v \neq 0$ is an *eigenvector* (or characteristic vector) of A with corresponding *eigenvalue* or (characteristic root) λ if

$$Av = \lambda v$$

To understand the geometry of a matrix A , we want to find a basis $\{v_i\}$ of the column space of A s.t.

$$Av_i = \lambda_i v_i$$

that is, a basis of eigenvectors. The intuition here is that if v_i lies on an axis in the column space of A , then applying A to v_i is only going to, at most, scale v_i . Identifying these axes and their associated scaling factor fully describes the geometry of the transformation that results from applying A . Note that

$$Av = \lambda v \iff (A - \lambda I)v = 0$$

Since $v \neq 0$,

$$p(\lambda) \equiv \det(A - \lambda I) = 0$$

$p(\lambda)$ is called the *characteristic polynomial* of A .³

Example 2. Consider

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \quad \det \begin{bmatrix} 1 - \lambda & 2 \\ 2 & 1 - \lambda \end{bmatrix} = 0 \iff (1 - \lambda)^2 - 4 = 0 \iff (\lambda + 1)(\lambda - 3) = 0$$

³On a technical note, the exposition here assumes that eigenvectors with real eigenvalues exist for A . To be more general with our exposition, we should expand our scalar field to $\mathbb{C}$ as it can be proven that matrices defined over $\mathbb{C}$ always have eigenvectors.

Hence $\lambda = \{-1, 3\}$. These are the *eigenvalues*. Now we find the *eigenvectors*. For $\lambda = -1$

$$(A - \lambda I)v = 0 \iff \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} v = 0$$

Which gives that v must be s.t. $v_1 = -v_2$. For $\lambda = 3$

$$(A - \lambda I)v = 0 \iff \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} v = 0$$

Which gives that v must be s.t. $v_1 = v_2$. It is often useful for the norm of the basis to be 1 where the norm $\|v\|$ for any $v \in \mathbb{R}^N$ is

$$\|v\| = \left(\sum_{i=1}^N v_i^2 \right)^{1/2}$$

Hence the normal eigenvectors are given by

$$v = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \text{and} \quad u = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Note that we can express the above as a system:

$$AP = P\Lambda$$

where

$$P = \begin{bmatrix} v & u \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \quad \Lambda = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}$$

In this case, note $v^T u = 0$ and $v^T v = u^T u = 1$. Hence

$$P^T P = \begin{bmatrix} v^T \\ u^T \end{bmatrix} \begin{bmatrix} v & u \end{bmatrix} = \begin{bmatrix} v^T v & v^T u \\ u^T v & u^T u \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

and if we pre-multiply the eigen-system by P^T we get

$$P^T AP = \Lambda$$

Definition 21. A an $N \times N$ matrix is *diagonalizable* if $\exists P$ and diagonal matrix Λ s.t.

$$P^{-1}AP = \Lambda$$

Note we can equivalently write $A = P\Lambda P^{-1}$.

If A is diagonalizable:

- A^m for any $m \in \mathbb{N}$ can be computed as $P\Lambda^m P^{-1}$.

$$A^m = A \times \dots \times A = P\Lambda P^{-1} \times \dots \times P\Lambda P^{-1} \times \dots \times P\Lambda P^{-1} = P\Lambda^m P^{-1}$$

since each of the $P^{-1}P$ cancel and $\Lambda \times \Lambda = \Lambda^2$ and so on.

- $A^{1/m}$ for any $m \in \mathbb{N}$ can be computed as $P\Lambda^{1/m} P^{-1}$.

$$\left(P\Lambda^{1/m} P^{-1}\right)^m = P\Lambda^{1/m} P^{-1} \times \dots \times P\Lambda^{1/m} P^{-1} = P\Lambda^{1/m} \times \dots \times \Lambda^{1/m} P^{-1} = P\Lambda P^{-1} = A$$

- If Λ is invertible, then A^{-m} for any $m \in \mathbb{N}$ can be computed as $P\Lambda^{-m} P^{-1}$.

$$A^{-m} = (A^m)^{-1} = (P\Lambda^m P^{-1})^{-1} = P\Lambda^{-m} P^{-1}$$

and we can check

$$A^m A^{-m} = P\Lambda^m P^{-1} P\Lambda^{-m} P^{-1} = P\Lambda^m \Lambda^{-m} P^{-1} = PP^{-1} = I$$

- Similarly, if Λ is invertible $A^{-1/m}$ for any $m \in \mathbb{N}$ can be computed as $P\Lambda^{-1/m} P^{-1}$.
- Let $q \in \mathbb{Q}_{\geq 0}$ (so $q = m/n$ for $m, n \in \mathbb{Z}$ with $n \neq 0$). Then $A^q = (A^m)^{1/n}$ is $P(\Lambda^m)^{1/n} P^{-1} = P\Lambda^q P^{-1}$.
- Finally, $A^r = P\Lambda^r P^{-1}$ for any $r \geq 0$ (if in addition Λ is invertible, this holds for any $r \in \mathbb{R}$).

Claim 2. *If A is diagonalizable then P is a matrix of eigenvectors and Λ is a matrix of eigenvalues.*

Proof. If A is diagonalizable then

$$AP = P\Lambda$$

Let $\mathbf{v}_k$ be the k th column of P . Then $A\mathbf{v}_k = \lambda_k \mathbf{v}_k$ we can see $\mathbf{v}_k$ is an eigenvector and λ_k an eigenvalue. $\square$

Theorem 6. *Let A be an $N \times N$ diagonalizable matrix with eigenvalues $\lambda_1, \dots, \lambda_N$.*

- $\text{trace}(A) = \sum_{i=1}^N \lambda_i$
- $\det(A) = \prod_{i=1}^N \lambda_i$.

Proof. Let us show this assuming A is diagonalizable.⁴

$$\begin{aligned}\text{trace}(A) &= \text{trace}(P^{-1}\Lambda P) = \text{trace}(\Lambda P P^{-1}) = \text{trace}(\Lambda) = \sum_i \lambda_i \\ \det(A) &= \det(P^{-1}\Lambda P) = \det(P^{-1}) \det(\Lambda) \det(P) = \det(\Lambda) = \prod_i \lambda_i\end{aligned}$$

where we utilized $\text{trace}(AB) = \text{trace}(BA)$ and $\det(A^{-1}) = \det(A)^{-1}$ □

Theorem 7. *If A is symmetric then A is diagonalizable and P can be chosen to be orthonormal ($P^{-1} = P^T$ with each column of P equal to a unit vector). This is known as the **spectral decomposition**.*

Recall in Example 2, the matrix A we considered was symmetric, and we found a diagonalization where $P^{-1} = P^T$. This theorem states that we can do this for any symmetric matrix.

Definition 22. The $N \times N$ matrix A is **positive definite** (**negative definite**) or simply PD (ND) if $\mathbf{v}'A\mathbf{v} > 0$ ($\mathbf{v}'A\mathbf{v} < 0$) for any nonzero vector $\mathbf{v} \in \mathbb{R}^N$. The matrix A is **positive semi-definite** (**negative semi-definite**) or simply PSD (NSD) if $\mathbf{v}'A\mathbf{v} \geq 0$ ($\mathbf{v}'A\mathbf{v} \leq 0$) for any $\mathbf{v} \in \mathbb{R}^N$.

Theorem 8. *Let A be a symmetric $N \times N$ matrix. Then*

1. A is PD (ND) $\iff \lambda_i > 0$ (< 0) for all eigenvalues λ_i of A .
2. A is PSD (NSD) $\iff \lambda_i \geq 0$ (≤ 0) for all eigenvalues λ_i of A .

Proof. If A is symmetric, then

$$\mathbf{v}'A\mathbf{v} = \mathbf{v}'P\Lambda P^T\mathbf{v} = \mathbf{u}'\Lambda\mathbf{u} \tag{2}$$

for $\mathbf{u} = P^T\mathbf{v}$ (since P is non-singular, $\mathbf{u} \neq 0$ if $\mathbf{v} \neq 0$). Hence the definiteness of A is the same as the definiteness of Λ . From here, we can see that

$$\mathbf{u}'\Lambda\mathbf{u} = \sum_{i=1}^N \lambda_i u_i^2 \tag{3}$$

where $u_i^2 \geq 0$ (strict for at least one i). If $\lambda_i > 0$ for all i , $\mathbf{v}'A\mathbf{v} = \sum_{i=1}^N \lambda_i u_i^2 > 0$, so it is PD. Conversely, if A is PD, we know $\mathbf{v}'A\mathbf{v} > 0$ for any $\mathbf{v} \neq 0$. In particular, it must be true for $\mathbf{v}$ equal to each of the columns of P . Since the columns of P are orthonormal, for $\mathbf{v} = \mathbf{p}_i$ the i th column of P , $P^T\mathbf{p}_i = \mathbf{e}_i$ the i th standard vector. Hence $\mathbf{v}'A\mathbf{v} = \lambda_i$ when $\mathbf{v} = \mathbf{p}_i$; since all such quadratic forms are positive, $\lambda_i > 0$. The logic for < 0 and ND, ≥ 0 and PSD, and ≤ 0 and NSD is completely analogous. □

⁴For the general proof, recall $\lambda_1, \dots, \lambda_N$ are the roots of the polynomial given by

$$\det(A - \lambda I) = (\lambda_1 - \lambda) \times (\lambda_2 - \lambda) \times \dots \times (\lambda_N - \lambda)$$

For the determinant, set $\lambda = 0$ and find $\det(A) = \prod_{i=1}^N \lambda_i$. For the trace, we need to leverage something known as “Vieta’s formulas.” The relevant result is that for a polynomial of order N , the $N - 1$ coefficient is the sum of the roots of the polynomial (this sounds esoteric, but think about how you learned to expand formulas like $(\lambda_1 - \lambda)(\lambda_2 - \lambda)$; the coefficient of the “middle” term is $-(\lambda_1 + \lambda_2)$, and this is just the generalization). If we can show $\text{trace}(A)$ is the $N - 1$ coefficient of the characteristic polynomial we’d be done. To see it, note that λ^{N-1} will only appear if all the diagonal elements are multiplied (any other permutation will give at most a polynomial of order $N - 2$). Hence λ^{N-1} only appears as part of the term

$$\prod_i (a_{ii} - \lambda)$$

Here the roots are a_{ii} , so the coefficient on λ^{N-1} is $\sum_i a_{ii} = \text{trace}(A)$.

5.1 Eigen Example: Eigen Decomposition of an Asymmetric Matrix

Let A be a 2×2 matrix (over $\mathbb{C}$) defined as follows

$$A = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$$

We can find that $\det(A - \lambda I) = 0$ gives

$$0 = (1 - \lambda)^2 + 4 = \lambda^2 - 2\lambda + 5 \quad \lambda = \frac{2 \pm \sqrt{4 - 4 \cdot 5}}{2}$$

Hence $\lambda = 1 \pm 2i$. We first consider the eigenvectors corresponding to the eigenvalue $1 + 2i$:

$$\begin{aligned} Av &= (1 + 2i)v \\ \iff \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} &= \begin{bmatrix} (1 + 2i)v_1 \\ (1 + 2i)v_2 \end{bmatrix} \\ \implies v_1 + 2v_2 &= (1 + 2i)v_1 \\ \implies v_2 &= iv_1 \end{aligned}$$

We can follow the same logic for $\lambda = 1 - 2i$ to arrive at the condition that defines the second set of eigenvectors:

$$\begin{aligned} Au &= (1 - 2i)u \\ \implies u_2 &= -iu_1 \end{aligned}$$

For simplicity, let $v_1 = u_1 = 1$. Then our eigenvectors are

$$v = \begin{bmatrix} 1 \\ i \end{bmatrix} \quad u = \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

Note that we could scale v and u to make them normal vectors; however, there is nothing we can do to make v and u orthogonal (i.e., $u'v = v'u = 0$). Thus, the matrix

$$P = \begin{bmatrix} v & u \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix}$$

is not an orthogonal matrix and instead of $P^{-1} = P^T$, we have that

$$P^{-1} = \begin{bmatrix} 1/2 & -1/2i \\ 1/2 & 1/2i \end{bmatrix}$$

Hence

$$A = \begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix} \begin{bmatrix} 1 + 2i & 0 \\ 0 & 1 - 2i \end{bmatrix} \begin{bmatrix} 1/2 & -1/2i \\ 1/2 & 1/2i \end{bmatrix}$$

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